



REAL TIME POWERLIFTING FORM ASSESSMENT USING YOLOv5 AND MEDIAPIPE

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ABSTRACT

This study introduces a real-time AI posture correction tool for bench press, squat, and deadlift, using YOLOv5 for object detection and MediaPipe for joint landmark extraction. Leveraging 3D joint coordinates from the PolyView Kinematic Corpus (PKC) dataset a custom collection of smartphone-captured powerlifting data from multiple angles the system trains models with our approach using traditional machine learning classifiers logistic regression, ridge classification, random forest, and gradient boosting to classify form and deliver phase-specific feedback. Results show this traditional ML method outperforms deep learning, providing precise guidance that reduces injury risk and boosts performance across diverse viewpoints.

INTRODUCTION

The COVID-19 pandemic spurred a global shift toward home workouts and virtual fitness apps like Peloton and Perfect Posture. While convenient, these solutions often lack real-time, exercise-specific form correction particularly for complex, high-risk movements such as the bench press, squat, and deadlift. Poor technique in these compound lifts can lead to serious injuries, yet most existing systems focus on general or single-joint exercises without phase-specific feedback. To bridge this gap, we introduce a real-time posture correction system that combines MediaPipe for landmark detection and YOLOv5 for exercise recognition. Powered by our custom PolyView Kinematic Dataset (PKC), which captures 3D joint coordinates across concentric and eccentric phases from multiple angles, the system provides detailed, real-time feedback to ensure safe and effective training.

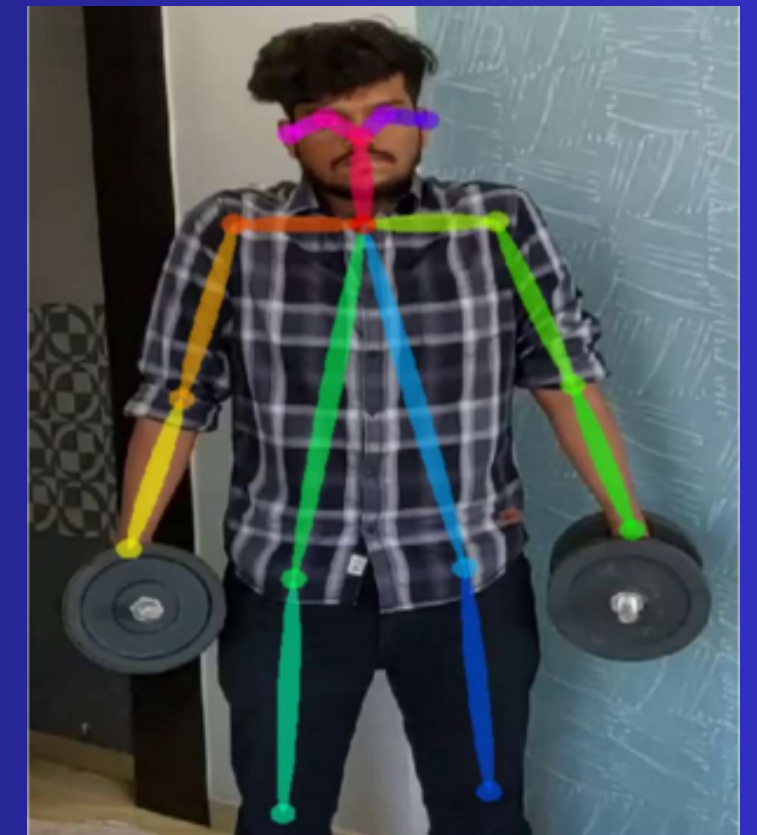
PAPER REFERENCE

Pose Estimation and Correcting Exercise Posture

https://www.itm-conferences.org/articles/itmconf/pdf/2021/05/itmconf_icacc2021_03031.pdf

EXISTING SYSTEM

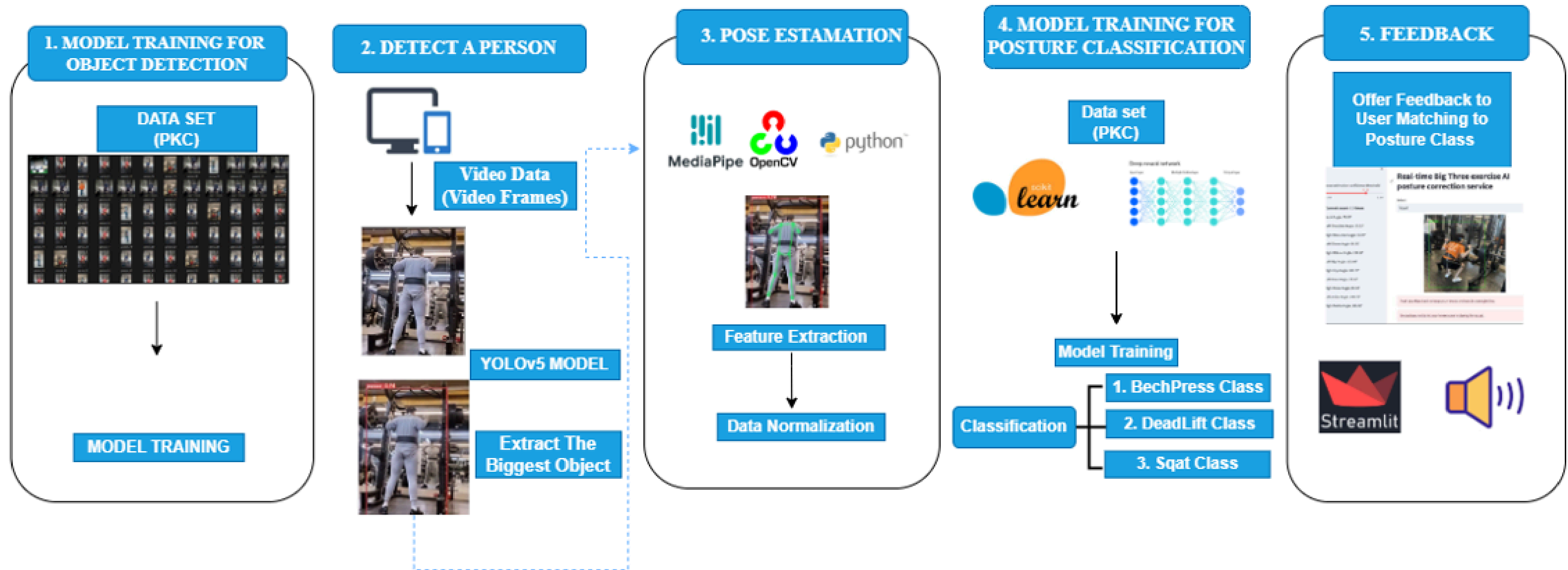
The Pose Trainer system captures RGB video at 720p and 30 FPS for exercises like squats and push-ups, targeting beginners to address form issues and injury risk, but its 2D joint detection limits depth error detection. Our proposed real-time powerlifting system improves accuracy with YOLOv5 and MediaPipe's 3D tracking at 30–35 FPS, offering live overlays, voice cues, and deeper insights via Streamlit with 120–150 ms latency, using the PolyView Kinematic Corpus (PKC) and cosine-similarity analysis (89–92.5% accuracy) for on-the-fly correction, surpassing Pose Trainer's 2D limitations.



PROPOSED SYSTEM

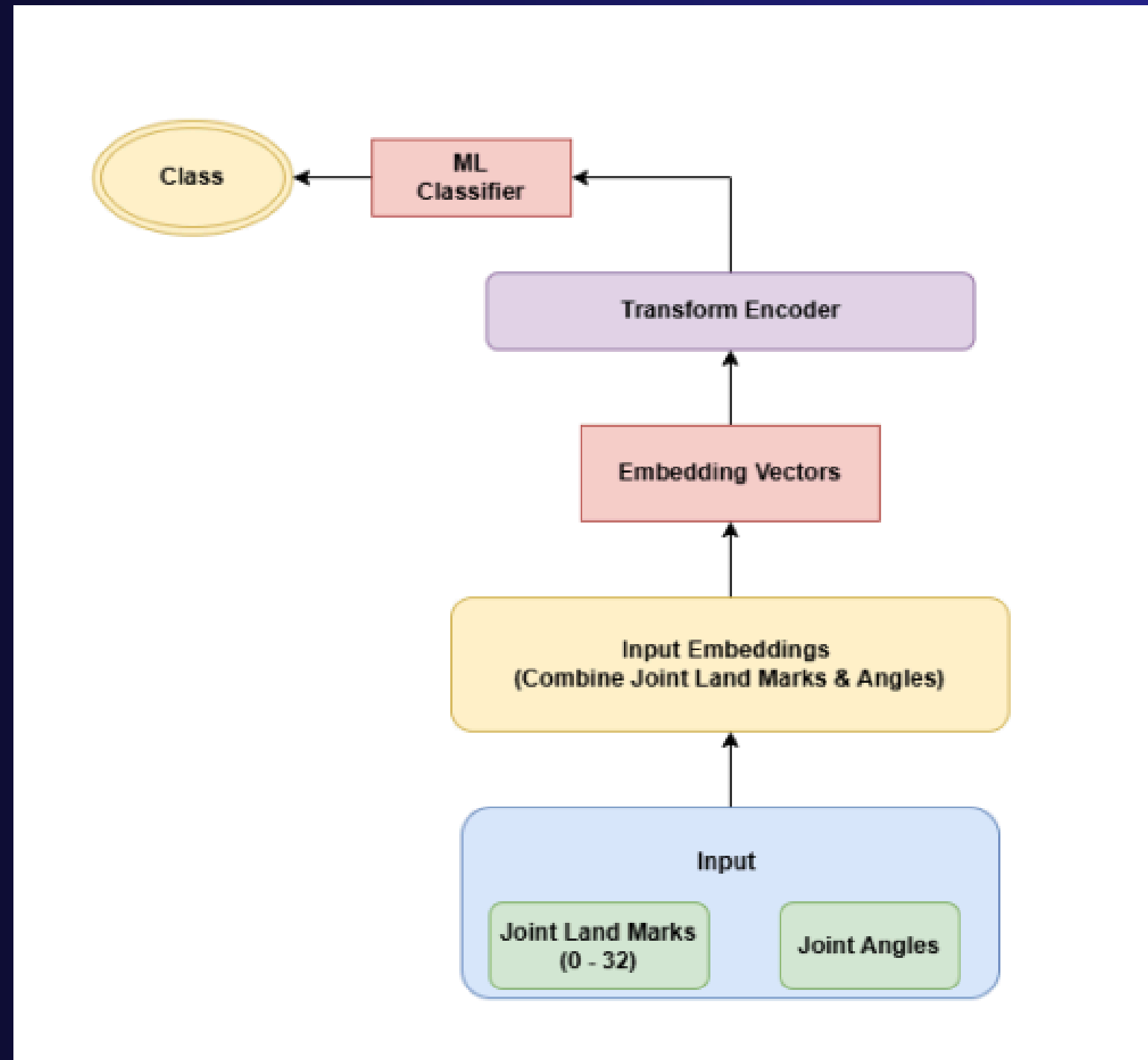
- Real-time AI system for posture correction in squats, bench press, and deadlifts.
- Exercise images collected and labeled from online sources.
- YOLOv5s used for detecting the person exercising via webcam.
- MediaPipe and OpenCV extract joint landmarks from video input.
- Joint angles are calculated, normalized, and classified using the PKC dataset.
- Streamlit-based web interface provides:
 - Real-time posture classification
 - Text and voice feedback
 - Joint angle data and rep count
- Helps users maintain correct posture and improve form.

PROPOSED SYSTEM



System WorkFlow

PROPOSED SYSTEM



Model Training Classifier Workflow

SOFTWARE REQUIREMENT SPECIFICATIONS

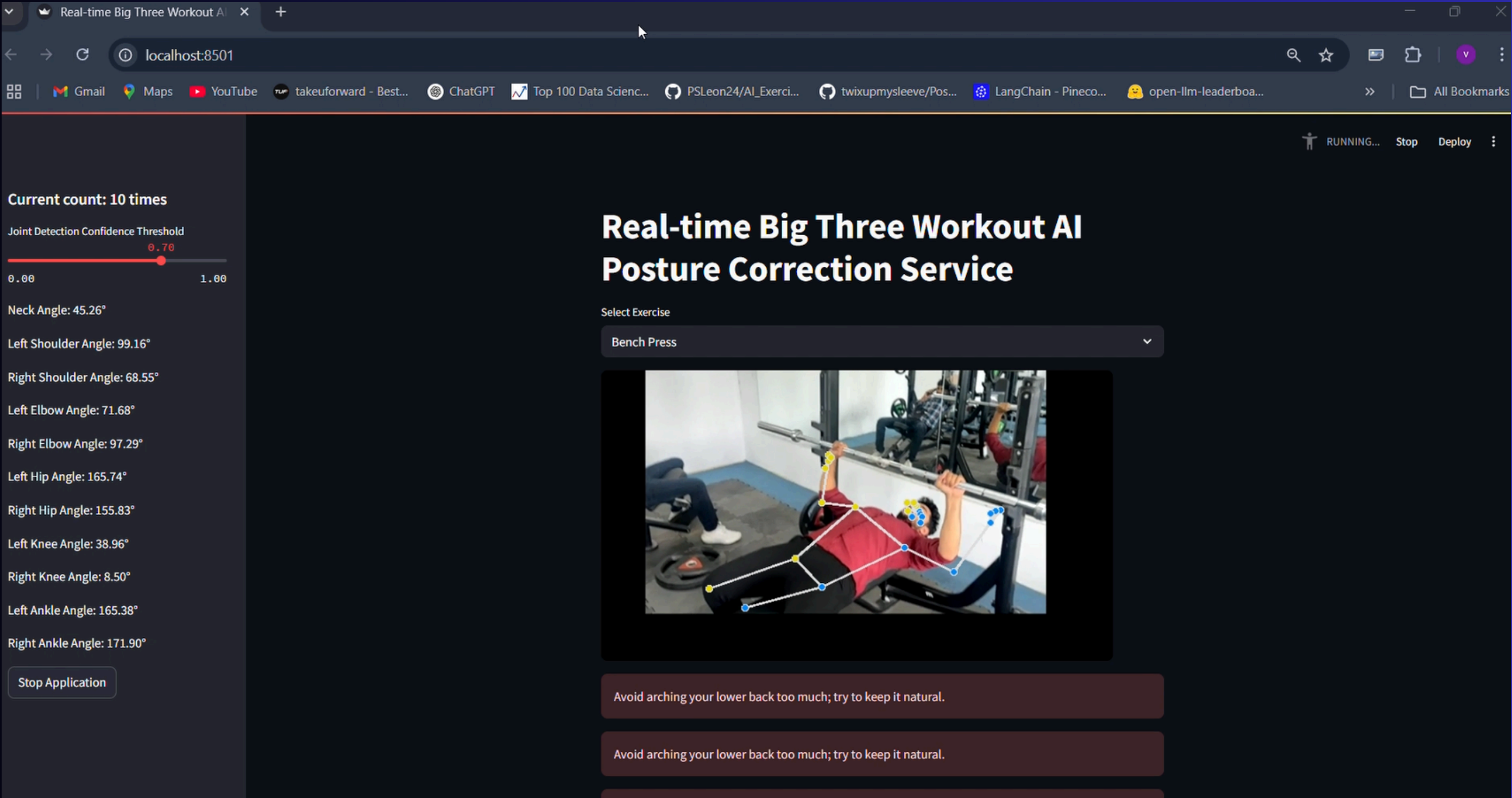
Hardware Requirements

- A PC or mobile device with a webcam
- Minimum RAM: 8 GB better to have 16 GB
- CPU: i5 or higher (for smooth video processing)
- GPU: It is recommended for faster model inference Minimum RTX 4060

Software Requirements

- Operating System: Windows/Linux/macOS
- Python (e.g., version 3.10+)
- Libraries: OpenCV, MediaPipe, YOLOv5, Streamlit, NumPy, etc.

OUTPUT SCREEN



Bench Press Feedback

OUTPUT SCREEN

Real-time Big Three Workout AI

localhost:8501

GmailMapsYouTubetakeuforward - Best...ChatGPTTop 100 Data Scienc...PSLeon24/AI_Exerci...twixupmysleeve/Pos...LangChain - Pineco...open-llm-leaderboa...

Real-time Big Three Workout AI Posture Correction Service

Select Exercise

Bench Press



Avoid arching your lower back too much; try to keep it natural.

Avoid arching your lower back too much; try to keep it natural.

Current count: 11 times

Joint Detection Confidence Threshold

0.76

0.001.00

Neck Angle: 50.57°

Left Shoulder Angle: 0.44°

Right Shoulder Angle: 1.37°

Left Elbow Angle: 134.52°

Right Elbow Angle: 177.45°

Left Hip Angle: 134.93°

Right Hip Angle: 87.08°

Left Knee Angle: 143.76°

Right Knee Angle: 79.02°

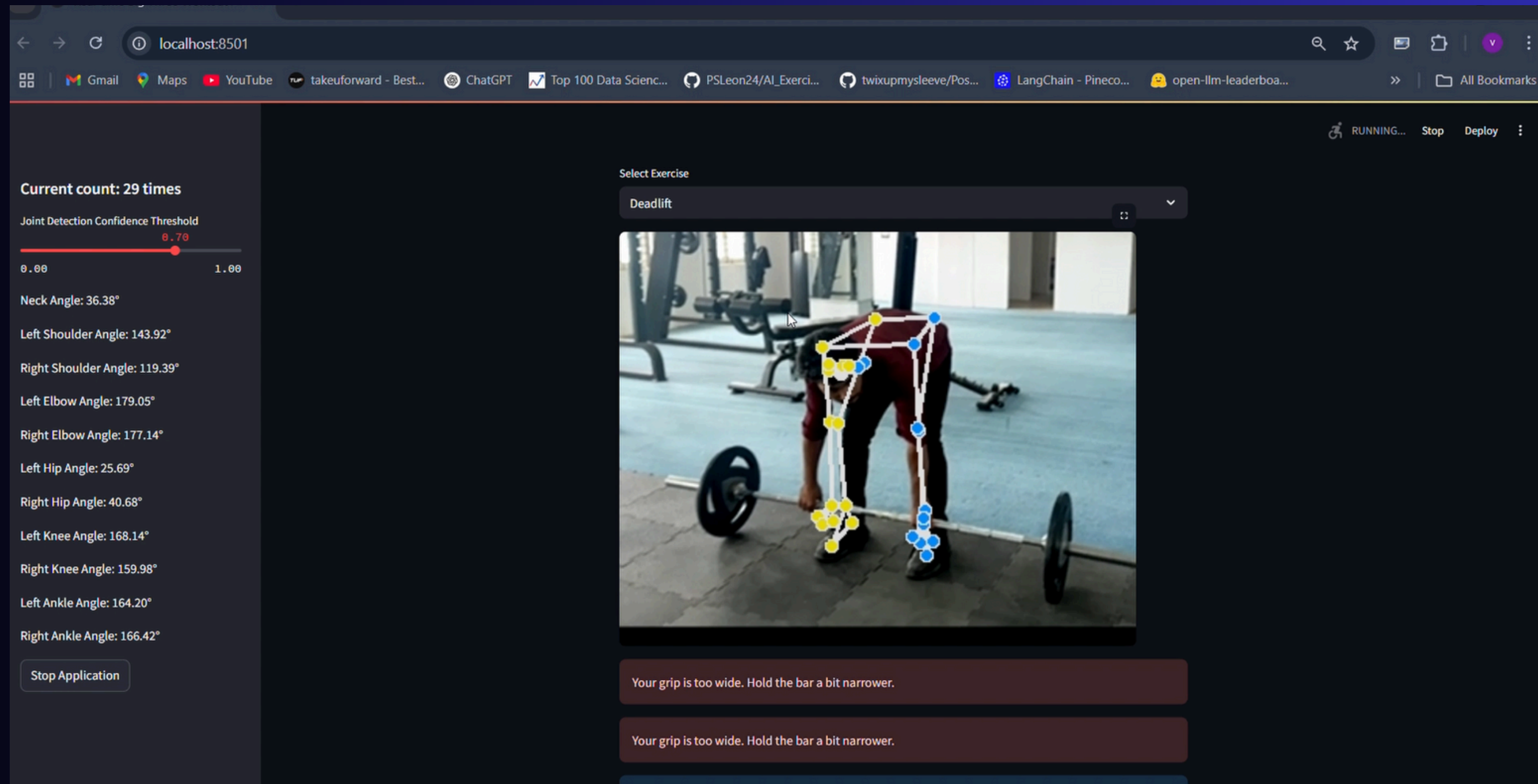
Left Ankle Angle: 174.21°

Right Ankle Angle: 153.32°

Stop Application

Sqat Feedback

OUTPUT SCREEN



DeadLift Feedback

CONCLUSION

In this work, we utilized MediaPipe's real-time pose estimation to extract joint landmarks and angles from the PolyView Kinematic Corpus (PKC) dataset, training traditional machine learning classifiers logistic regression, ridge classification, random forest, and gradient boosting to distinguish correct and incorrect forms for bench press, squat, and deadlift. The models were deployed in a web browser using OpenCV for video processing, Streamlit for an interactive interface, and YOLOv5 for precise exerciser detection, analyzing frames with high accuracy and delivering text-to-speech feedback on standard CPU and GPU hardware. By focusing on traditional classifiers, our lightweight, interpretable approach provides frame-by-frame corrections, laying a robust foundation for practical powerlifting use.

THANK YOU